Land Acquisition Workflow

A Proprietary P2P/Escrow Interface

Bridging Blockchain Logic with Real-World Legal Milestones

System Architecture and Implementation Guide

Platform Features

- Multi-Signature Wallet Security
- Notary-Mediated Escrow System
- 4-Phase Workflow Implementation
- Document Hashing & Verification
- Fiat Bridge Integration (Crypto to KES)
- Smart Contract Automation

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1. Executive Summary

1.1. Key Innovation

This platform represents a revolutionary approach to land acquisition, combining blockchain technology with traditional legal processes. By integrating Ethereum smart contracts with real-world legal milestones, we create a trustless, transparent, and secure environment for property transactions.

1.2. Core Components

- **MultiSigWallet:** A secure multi-signature wallet requiring M-of-N confirmations
- **NotaryMediatedEscrow:** State machine managing the entire transaction lifecycle
- **EscrowFactory:** Factory contract for deploying new escrow instances

2. The 4-Phase Workflow

2.1. Phase 1: Intake & Discovery

The process begins when a client submits their requirements:

1. Client creates an intake with budget, desired property size, and purpose
2. Notary uploads digital gallery of available properties
3. Client selects a property from the discovery phase
4. Client funds the escrow with due diligence fee

Key Features:

- Budget validation against minimum due diligence fee
- Digital gallery for property exploration
- Property selection triggers plot locking

2.2. Phase 2: Due Diligence "Safety Gate"

This critical phase involves comprehensive verification:

1. Notary conducts registry search
2. Notary performs site visit
3. Notary uploads due diligence report
4. Client verifies and releases due diligence milestone payment

Security Features:

- Title search verification
- Site inspection reports
- Document hashing for immutability
- Multi-party verification requirements

2.3. Phase 3: Execution & The Fiat Bridge

Once due diligence passes, execution begins:

1. Both parties sign the sale agreement on-platform
2. Notary notarizes the signed document
3. Client deposits full remaining amount
4. Fiat bridge converts crypto to KES (K Kenyan Shilling)
5. Funds disbursed to seller (minus facilitation fee)

Fiat Bridge Features:

- Notary executes fiat bridge with exchange rate
- Converts cryptocurrency to local currency
- Records KES amount and exchange rate on-chain
- Disbursement to seller and facilitation fee to treasury

2.4. Phase 4: Transfer Registry & Finalization

The final phase handles title transfer:

1. Notary uploads lodgement receipt
2. Client confirms final transfer
3. Notary uploads original title documents
4. Multi-sig owners confirm final release
5. Title issued with courier tracking

Completion Features:

- Original title document verification
- Land Registry confirmation
- Courier tracking integration
- Facilitation fee release to treasury

3. Smart Contract Architecture

3.1. MultiSigWallet.sol

The MultiSigWallet provides secure fund management:

```
contract MultiSigWallet {
    address[] public owners;
    mapping(address => bool) public isOwner;
    uint public numConfirmationsRequired;

    struct Transaction {
        address to;
        uint value;
        bytes data;
        bool executed;
        uint numConfirmations;
    }
}
```

Key Features:

- M-of-N signature requirement
- Transaction submission by owners
- Confirmation and execution flow
- Non-reentrant protection

3.2. NotaryMediatedEscrow.sol

The core escrow state machine:

```
enum EscrowState {
    Created, Intake, Discovery, PlotSelected,
    Funded, DueDiligencePending, DueDiligenceComplete,
    ExecutionPending, ExecutionComplete,
    TransferRegistry, Completed, Cancelled
}
```

State Transitions:

1. Intake → Discovery (gallery upload)
2. Discovery → PlotSelected (plot selection)
3. PlotSelected → DueDiligencePending (funding)
4. DueDiligencePending → DueDiligenceComplete (DD completion)
5. DueDiligenceComplete → ExecutionPending (proceed)
6. ExecutionPending → ExecutionComplete (fiat bridge)

7. ExecutionComplete → TransferRegistry (lodgement)
8. TransferRegistry → Completed (final transfer)

3.3. EscrowFactory.sol

Factory pattern for creating escrows:

```
function createEscrow(
    address _seller,
    address _notary,
    uint _totalAmount
) public returns (address escrowContract)
```

Features:

- Default notary and owners configuration
- Batch escrow creation
- Treasury and fee configuration
- Pending action timelock mechanism

4. Financial Flow

4.1. Fee Structure

Fee Type	Default	Max
Due Diligence Fee	0.05 ETH	-
Facilitation Fee	2%	5%
Deposit	10%	Variable

Notary can negotiate rates after due diligence completion.

4.2. Release Schedule

1. Due Diligence: $200 to Notary
2. Deposit: To seller upon execution
3. Balance: To seller minus facilitation fee
4. Facilitation Fee: To treasury

4.3. Treasury Configuration

The treasury receives:

- Facilitation fee (configurable, max 5%)
- Any cancelled escrow deposits

5. Notary Admin Panel Features

5.1. Dashboard

Central hub for notarization workflow:

- Pending intakes overview
- Active escrow status
- Due diligence tracking
- Document verification queue

5.2. Fiat Payment Logging

- KES amount entry
- Exchange rate recording
- Conversion verification
- On-chain event emission

5.3. Timeline Management

- Expected completion dates
- Milestone tracking
- Automated notifications
- Progress visualization

5.4. Document Management Hub

Secure section for legal documents:

- Search certificates
- Due diligence reports
- Sale agreements
- Title documents
- LCB Consents

Document Hashing:

- SHA-256 hash stored on-chain
- Client verification capability
- Immutable audit trail

6. Security Features

6.1. Multi-Signature Verification

- All fund releases require multi-sig approval
- Configurable M-of-N threshold
- Separate execution from confirmation

6.2. Document Hashing & Verification

- All documents hashed using keccak256
- Hashes stored in contract state
- Client can verify document authenticity

6.3. Access Control

Role-based permissions:

- Client: Create intake, sign documents, verify
- Notary: Upload reports, execute fiat bridge, complete transfer
- Treasury: Upload gallery, receive facilitation fees
- Owners: Confirm and execute transactions

6.4. State Machine Protection

- Strict state transition validation
- No skipping phases
- Reentrancy guards on critical functions

7. Technical Implementation

7.1. Technology Stack

- **Smart Contracts:** Solidity 0.8.19
- **Framework:** Hardhat
- **Frontend:** React + Ethers.js
- **Deployment:** Localhost/Hardhat node

7.2. Deployment Process

1. Deploy MultiSigWallet with owners
2. Deploy EscrowFactory with treasury and notary
3. Configure default owners and confirmations
4. Set up frontend with contract addresses

7.3. Network Configuration

- Localhost: 127.0.0.1:8545 (Chain ID: 1337)
- Browser wallet (MetaMask) connection

Environment Variables:

```
MULTISIG_ADDRESS=0x...
ESCROW_FACTORY_ADDRESS=0x...
LEGAL_NOTARY_ADDRESS=0x...
TREASURY_ADDRESS=0x...
```

8. Operational Guidelines

8.1. Client Onboarding

1. Connect wallet to platform
2. Submit intake (budget, size, purpose)
3. Review digital gallery
4. Select property
5. Fund due diligence fee
6. Verify due diligence report
7. Sign sale agreement
8. Fund remaining amount
9. Confirm final transfer

8.2. Notary Responsibilities

1. Review and approve intake requests
2. Upload property gallery
3. Conduct registry search
4. Perform site visit
5. Upload due diligence report
6. Execute fiat bridge
7. Upload lodgement receipt
8. Complete final transfer

8.3. Admin Team Workflow

1. Monitor active escrows
2. Process escalated issues
3. Manage treasury operations
4. Configure fee structures

9. Compliance & Legal

9.1. Regulatory Considerations

- KYC/AML compliance handled off-chain
- Notary acts as regulated entity
- Document retention policies apply

9.2. Document Retention

- All hashes stored permanently
- Original documents off-chain
- Verification available indefinitely

9.3. Dispute Resolution

- Multi-sig provides dispute resolution mechanism
- Notary serves as mediator
- Smart contract state serves as evidence

10. Future Enhancements

10.1. Planned Features

- IPFS integration for document storage
- Oracle integration for price feeds
- Additional fiat bridge currencies
- Fractional property ownership

10.2. Scalability Considerations

- Layer 2 scaling solutions
- Batch transaction processing
- Optimized gas usage

11. Conclusion

This platform represents a significant advancement in secure, transparent land acquisition. By combining blockchain security with traditional legal processes, we create a trustless environment that benefits all parties.

The 4-phase workflow ensures proper verification at each step, while the multi-signature wallet protects all funds. The notary-mediated approach adds human verification to automated smart contract logic.

Future enhancements will expand capabilities while maintaining the core security principles established in this implementation.

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